

BROWN FIELD, CA, USA

WMO: 722904

Lat: 32.572N

Lon: 116.979W

Elev: 515

StdP: 14.42

Time Zone: -8.00 (NAP)

Period: 97-19

WBAN: 03178

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	39.1	41.8	7.9	8.4	61.8	13.2	10.9	62.2	20.5	61.1	18.3	60.5	6.0	120	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.8	89.9	64.3	85.8	64.6	82.4	64.8	71.5	82.3	70.1	79.9	68.9	77.8	8.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	105.0	75.3	66.5	99.5	74.2	65.6	96.3	73.5	35.5	82.5	34.4	79.9	33.3	78.1	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.2	12.9	11.9	DB	34.6	98.3	2.4	3.5	32.9	100.8	31.5	102.8	30.1	104.8	28.4	107.4	
			WB	29.7	73.5	2.5	1.9	27.9	74.9	26.4	76.0	25.0	77.0	23.2	78.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	56.2	55.5	57.7	59.8	62.8	66.0	70.0	71.0	70.5	66.3	60.9	55.6
	DBStd	7.34	5.58	5.18	4.67	4.70	3.98	3.45	3.63	3.61	4.80	5.09	5.63	4.90
	HDD50	23	8	6	1	0	0	0	0	0	0	0	1	7
	HDD65	1557	279	269	235	169	91	23	1	1	4	41	152	292
	CDD50	4669	199	162	239	294	397	479	619	652	614	505	327	182
	CDD65	730	6	5	8	13	23	52	155	188	169	81	28	2
	CDH74	4630	99	88	136	206	220	209	661	914	1028	696	329	44
CDH80	1354	15	19	41	69	84	53	133	216	338	271	111	4	
Wind	WSAvg	5.2	5.4	5.9	5.3	5.8	5.6	5.1	4.7	4.5	4.5	4.8	5.1	5.5
Precipitation	PrecAvg	12.0	1.9	2.5	1.7	1.0	0.3	0.0	0.1	0.1	0.2	0.6	1.2	2.0
	PrecMax	22.4	5.0	10.1	3.7	2.0	1.6	0.2	0.9	0.2	1.2	5.6	3.5	6.3
	PrecMin	6.4	0.2	0.1	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.1
	PrecStd	4.3	1.6	2.3	0.9	0.6	0.4	0.0	0.2	0.1	0.3	1.3	0.6	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.0	82.8	85.2	87.9	90.0	86.9	88.2	90.7	95.1	93.8	89.1	79.1
		MCWB	55.4	56.1	57.8	57.2	59.4	67.0	70.4	69.3	67.8	61.8	57.6	53.5
	2%	DB	76.5	75.9	77.1	80.1	80.1	79.0	83.5	85.3	88.1	87.5	83.0	73.5
		MCWB	53.8	54.6	56.2	57.8	61.7	66.6	69.6	69.5	68.7	60.2	56.7	52.3
	5%	DB	72.2	70.4	71.7	73.5	73.4	75.2	80.4	81.8	83.5	81.6	76.9	70.0
		MCWB	52.7	53.4	56.2	57.6	61.4	64.8	68.4	69.1	68.2	60.3	55.0	52.0
	10%	DB	67.3	65.4	67.0	69.4	70.4	72.9	77.4	79.1	80.0	76.4	71.9	65.6
		MCWB	51.9	53.1	56.1	58.0	60.5	63.5	67.3	68.1	67.3	60.6	54.6	52.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	60.4	62.2	63.1	66.5	69.3	72.9	72.9	73.5	69.4	63.3	61.2
		MCDB	68.4	67.7	74.2	75.5	77.9	82.2	82.8	85.1	86.6	77.4	72.7	67.3
	2%	WB	58.4	58.2	60.1	61.4	64.2	67.0	70.7	71.0	71.7	66.8	61.3	58.8
		MCDB	65.9	66.8	69.7	72.6	74.2	77.8	80.6	82.1	82.7	75.9	70.6	64.3
	5%	WB	56.9	56.8	58.7	59.8	62.4	65.6	69.2	69.5	70.1	65.0	59.8	57.2
		MCDB	64.1	65.2	67.2	69.3	71.4	74.4	78.5	79.4	78.8	73.8	68.3	62.5
	10%	WB	55.4	55.3	57.3	58.4	60.9	64.2	68.0	68.5	68.6	63.5	58.6	55.5
		MCDB	62.4	63.4	64.6	66.7	68.8	72.1	76.2	77.4	76.4	71.3	66.6	61.2
Mean Daily Temperature Range	5% DB	MDBR	20.5	19.0	18.6	18.0	15.2	14.4	15.0	15.8	17.1	18.9	20.6	20.2
		MCDBR	27.7	28.1	28.1	27.3	22.0	17.8	18.5	19.7	22.9	28.7	28.8	26.8
		MCWBR	14.0	14.1	14.0	13.3	10.2	7.8	7.3	7.6	9.3	14.0	14.1	14.4
	5% WB	MCDBR	19.4	20.8	20.8	20.5	18.8	17.5	17.1	17.9	19.0	20.0	20.9	18.2
MCWBR		11.9	12.0	11.5	10.6	8.9	8.0	6.9	7.1	7.8	10.3	12.0	11.3	
Clear-Sky Solar Irradiance	taub		0.310	0.321	0.332	0.344	0.363	0.365	0.394	0.396	0.359	0.365	0.320	0.322
	taud		2.556	2.539	2.491	2.462	2.428	2.424	2.429	2.402	2.483	2.437	2.554	2.549
	Ebn at Noon		291	298	301	300	293	290	282	280	288	277	285	279
	Edh at Noon		26	29	33	35	37	37	36	37	33	32	26	25
All-Sky Solar Radiation	RadAvg		1063	1335	1753	2117	2235	2339	2249	2161	1883	1479	1153	963
	RadStd		66	111	97	109	162	134	147	99	101	105	72	66

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (33 neighbors)	+1.53	N/A	N/A	+2.59	+1.37	+1.26	N/A	-342	+565	+252
	+1.19	N/A	-2.90	+1.39	N/A	N/A	N/A	-191	+419	+227

Nomenclature: See separate page